David Yallup

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Research Themes: ***MCMC*** · ***Nested Sampling*** · ***GPU-Accelerated Inference*** · ***Generative AI for Science***

Profile

Postdoctoral researcher in probabilistic machine learning, foundational topics in learning theory and scientific inference. Specialising in GPU-accelerated Bayesian computation and generative surrogate modelling for physical simulations. Lead developer and contributor to open-source JAX scientific software, with experience translating research methods into robust, reusable code and collaborating across academia and industry.

Experience

Current appointments

Kavli Institute for Cosmology, University of Cambridge
Postdoctoral Research Associate *2021–*
Member of the Handley-lab research group:
- Developing GPU-accelerated inference for next-generation cosmology, gravitational wave, and particle physics experiments
- Building simulation-based inference and advanced MCMC tools in JAX/BlackJAX
- Bayesian statistics for scientific inference problems

Corpus Christi College, University of Cambridge
Research Associate *2022–*

Polychord Ltd.
Research Scientist *2021–*
Cambridge spin-out applying Bayesian methods to industrial problems including:
- Molecular structure prediction, battery design optimisation, Bayesian experimental design

Past positions

University College London
Postdoctoral Research Associate *2019–2020*
Development of statistical and ML inference methods for particle phenomenology

Simcorp Ltd.
Business Consultant *2013-2015*
Software development and consultancy for asset management industry clients

Education

UCL
PhD Particle Physics *2015–2019*
Thesis: *"Constraining new physics with fiducial LHC measurements,"* supervised by Prof J. Butterworth.
Recipient of UCL HEP postgraduate prize.

Durham University
MSc Particles, Strings and Cosmology *2014–2015*

Durham University
MSci Natural Sciences, Maths and Physics *2009–2013*

Grants

UKRI AIRR Gateway Award: Real-time gravitational wave inference
£50k Isambard-AI GPUh credits *2026*

Google Cloud for Researchers
$5k Google Compute Engine GPU credits *2025*

Kavli Foundation
£3k to host Cosmological Inference in High Dimensions workshop *2024*

Corpus Christi College, University of Cambridge
College Research Associateship *2023*

Marie Curie Early Career Researcher
£30k for academic secondment and travel grant *2016–2019*

Supervision

Part II Statistical Physics **Institute of Astronomy (Maths)**
University of Cambridge *2025*
4 groups of 2 students, ∼40 hours

Part II Relativity **Natural Sciences Tripos (Physics)**
University of Cambridge *2022*
4 groups of 3 students, ∼40 hours

Part III Projects **Natural Sciences Tripos (Physics)**
University of Cambridge *2021–*
Primary supervisor for 5 MSc projects, ∼20 hours contact each

MSci Project Supervision **Physics and Astronomy**
UCL *2016–2019*
Co-supervisor for 1 MSc project and 8 MSci students with Prof Butterworth

Selected Talks (2024–)

NeurIPS – FPI Workshop
San Diego *2025*
Particle Monte Carlo methods for lattice field theory

CamGW: Cambridge Gravitational Wave Network
KICC, University of Cambridge *2025*
Towards real-time gravitational wave inference

Future of SED Fitting
KICC, University of Cambridge *2025*
GPU-accelerated sampling methods

ICLR – FPI Workshop
Singapore *2025*
Nested slice sampling

Gravitational Wave Analysis in the Era of Machine Learning
Royal Astronomical Society *2025*
GPU-accelerated sampling methods

BayesAI Workshop
Lancaster University *2024*
Neural network advances in nested sampling

EU AI for Fundamental Physics Conference
Amsterdam *2024*
Diffusion meets nested sampling

Cavendish Astrophysics Seminar
University of Cambridge *2024*
Diffusion models for accelerated inference

Astrophysics ML Seminar
University of Cambridge *2024*
Simulation-based inference

Notable publications ⋆

[1] T. Lovick, **Yallup, David** and W. Handley, *Automatic laplace collapsed sampling: Scalable marginalisation of latent parameters via automatic differentiation*, 2026.

[2] D.D.Y. Ong, **Yallup, David** and W. Handley, *The Bayesian view of DESI DR2: Evidence and tension in a combined analysis with CMB and supernovae across cosmological models*, 2603.05472.

[3] **Yallup, David**, *Nested sampling with slice-within-gibbs: Efficient evidence calculation for hierarchical bayesian models*, 2026.

[4] N. Kroupa, **Yallup, David** and W. Handley, *Time delays and stationarity in quasar light curves*, *Phys. Rev. D* (2026) [2602.11264].

[5] **Yallup, David**, N. Kroupa and W. Handley, *Nested slice sampling: Vectorized nested sampling for gpu-accelerated inference*, 2026.

[6] **Yallup, David**, *Particle Monte Carlo methods for Lattice Field Theory*, in *39th Annual Conference on Neural Information Processing Systems: Includes Machine Learning and the Physical Sciences (ML4PS)*, 11, 2025 [2511.15196].

[7] D.D.Y. Ong, **Yallup, David** and W. Handley, *A Bayesian Perspective on Evidence for Evolving Dark Energy*, 2511.10631.

[8] **Yallup, David**, M. Prathaban, J. Alvey and W. Handley, *Parallel Nested Slice Sampling for Gravitational Wave Parameter Estimation*, in *2nd European AI for Fundamental Physics Conference*, 9, 2025 [2509.24949].

[9] A.N. Ormondroyd, W.J. Handley, M.P. Hobson, A.N. Lasenby and **Yallup, David**, *Dynamic or Systematic? Bayesian model selection between dark energy and supernova biases*, 2509.13220.

[10] T. Lovick, **Yallup, David**, D. Piras, A. Spurio Mancini and W. Handley, *High-Dimensional Bayesian Model Comparison in Cosmology with GPU-accelerated Nested Sampling and Neural Emulators*, 2509.13307.

[11] M. Prathaban, **Yallup, David**, J. Alvey, M. Yang, W. Templeton and W. Handley, *Gravitational-wave inference at GPU speed: A bilby-like nested sampling kernel within blackjax-ns*, 2509.04336.

[12] **Yallup, David**, N. Kroupa and W. Handley, *Nested slice sampling*, in *The Thirteenth International Conference on Learning Representations: Frontiers in Probabilistic Inference Workshop*, 2025, https://openreview.net/forum?id=ekbkMSuPo4.

[13] N. Kroupa, **Yallup, David**, W. Handley and M. Hobson, *Kernel-, mean-, and noise-marginalized Gaussian processes for exoplanet transits and H0 inference*, *Mon. Not. Roy. Astron. Soc.* **528** (2024) 1232 [2311.04153].

[14] **Yallup, David** and W. Handley, *Hunting for bumps in the margins*, *JINST* **18** (2023) P05014 [2211.10391].

[15] **Yallup, David**, W. Handley, M. Hobson, A. Lasenby and P. Lemos, *Split personalities in Bayesian Neural Networks: the case for full marginalisation*, 2205.11151.

[16] **Yallup, David**, Janßen, Timo, Schumann, Steffen and Handley, Will, *Exploring phase space with nested sampling*, *Eur. Phys. J. C* **82** (2022) 678.

[17] P. Lemos, M. Cranmer, M. Abidi, C. Hahn, M. Eickenberg, E. Massara et al., *Robust Simulation-Based Inference in Cosmology with Bayesian Neural Networks*, in *39th International Conference on Machine Learning Conference*, 7, 2022 [2207.08435].

[18] A. Buckley et al., *Testing new physics models with global comparisons to collider measurements: the Contur toolkit*, *SciPost Phys. Core* **4** (2021) 013 [2102.04377].

[19] S. Amrith, J. Butterworth, F. Deppisch, W. Liu and **Yallup, David**, *LHC Constraints on a $B-L$ Gauge Model using Contur*, *JHEP* **05** (2019) 154 [1811.11452].

[20] G. Brooijmans et al., *Les Houches 2017: Physics at TeV Colliders New Physics Working Group Report*, in *10th Les Houches Workshop on Physics at TeV Colliders*, 3, 2018 [1803.10379].

[21] J.M. Butterworth, D. Grellscheid, M. Krämer, B. Sarrazin and **Yallup, David**, *Constraining new physics with collider measurements of Standard Model signatures*, *JHEP* **03** (2017) 078 [1606.05296].

⋆ *As an ATLAS collaboration author I was an author on over 280 collaboration papers; only external small-author-list papers are listed here.*